Potentially new constructions

Low rank Grothendieck rings · Novelty shortlist and unresolved cases · 20 October 2026

To the best of our current knowledge, these seven constructions are promising candidates for new tensor categories. Their novelty up to tensor equivalence remains subject to independent verification.

Reported constructions to examine for novelty

m denotes maximum fusion multiplicity. Click a census ID for its construction source.

R6_4_249 Rank 6 · m = 4	**Sign-swap equivariantization** of the diagonal-fermion quotient of two identical half-E_6 categories. Known inputs: Aasen-Lake-Walker.
R6_4_322 Rank 6 · m = 4	**Mixed diagonal-fermion quotient** of Ad(E_6) and PSU(2)$_6$. Both input categories are already known.
R6_6_1712 Rank 6 · m = 6	**Relative Fibonacci center of TT$_3$.** The input TT$_3$ is the known dual even part of 2D2; see Edie-Michell, Definition 2.5.
R7_4_35 Rank 7 · m = 4	**Inversion descent and reflection Morita dual** of an orbifold of NG($C_4 \times C_4$,16). Input: Yu-Zhang.
R7_4_2665 Rank 7 · m = 4	**Analogous inversion/Morita construction** from NG(C_{16},16), with different invertibles and duality. Input: Budinski (2021).
R8_2_37 Rank 8 · m = 2	**Nonsplit C_2-extension of 2D2,** with an odd invertible a satisfying $a^2 = g$. Uses Grossman-Izumi-Snyder and Grossman.
R8_2_575 Rank 8 · m = 2	**Explicit Davydov-Runkel extension** of Kashina's self-dual 16-dimensional Hopf algebra. Simple dimensions: (1,1,1,1,2,2,2,4).

Known model added in the rank-seven extension

R7_5_5678 · Rank 7 · m = 5. Rep(AGL(1,7)) = Rep($C_7 \rtimes C_6$), of global dimension 42. The full fusion tensor is verified. This is a known representation category, not an additional novelty candidate.

OPEN: 4 fusion rings with no established categorification

R6_4_155 · Rank 6 · multiplicity 4. Generalized near-group ring R′(4): four invertibles forming $C_2 \times C_2$, and two mutually dual noninvertibles of dimension $4 + 3\sqrt{2}$. The ring already appears in Meng-Yu, Theorem 1.1.

R6_8_1660 · Rank 6 · multiplicity 8. Self-dual ring containing an Ising fusion subring. A further simple X has dimension $3 + 2\sqrt{2} + 2\sqrt{3} + \sqrt{6}$ and satisfies $X^2 = 1 + [3(1 + g) + 4\sigma]X$, where $\sigma^2 = 1 + g$.

Rank 7 · multiplicity 5: 2 unresolved rings. Of all 5,805 new rings, 5,802 are excluded and one is realized. The remaining census IDs are:

R7_5_3895 · R7_5_5039

All 4 remaining cases are open for general complex categorification; no positive model or complete exclusion is established. Exact fusion data.

AI-generated research · independent verification pending.
Independent human verification and Lean formalization remain pending. Please treat the claims with caution.